

Problem

Realize an LC ladder network such that

$$y_{22} = \frac{s^3 + 5s}{s^4 + 10s^2 + 8}$$

Step-by-step solution

Step 1 of 5

Consider the following admittance of an LC ladder network.

$$y_{22} = \frac{s^3 + 5s}{s^4 + 10s^2 + 8}$$

The order of the network is 4.

The output driving-point (input port short-circuited) admittance of the LC network is Y_{22} .

Determine the values of L and C that results the admittance Y_{22} .

Assume that the impedance of the network is Z_A .

The following is the impedance Z_A of LC ladder network:

$$\begin{aligned} Z_A &= \frac{1}{Y_{22}} \\ &= \frac{1}{\left(\frac{s^3 + 5s}{s^4 + 10s^2 + 8} \right)} \\ &= \frac{s^4 + 10s^2 + 8}{s^3 + 5s} \end{aligned}$$

Step 2 of 5

Draw the model of LC network by short circuiting the input port:

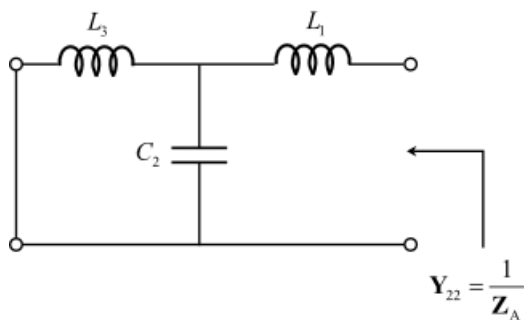


Figure 1

Step 3 of 5

The simplified expression for $\mathbf{Z}_A$ equals $sL_1 + \mathbf{Z}_B$.

Simplify the fraction $\mathbf{Z}_A$ using long division:

$$\frac{s^3 + 5s}{5s^2 + 8} s^4 + 10s^2 + 8 \left(s \right.$$

The following is the simplified expression for $\mathbf{Z}_A$.

$$\mathbf{Z}_A = s + \frac{5s^2 + 8}{s^3 + 5s}$$

Therefore,

$$L_1 = 1 \text{ H}$$

$$\mathbf{Z}_B = \frac{5s^2 + 8}{s^3 + 5s}$$

The following is the admittance corresponds to $\mathbf{Z}_B$.

$$\begin{aligned} \mathbf{Y}_B &= \frac{1}{\mathbf{Z}_B} \\ &= \frac{1}{\left(\frac{5s^2 + 8}{s^3 + 5s} \right)} \\ &= \frac{s^3 + 5s}{5s^2 + 8} \end{aligned}$$

Step 4 of 5

The simplified expression for $\mathbf{Y}_B$ equals $sC_2 + \mathbf{Y}_C$.

Simplify the fraction $\mathbf{Y}_B$ using long division:

$$\begin{array}{r} 5s^2 + 8 \big) s^3 + 5s \big(0.2s \\ \underline{s^3 + 1.6s} \\ 3.4s \end{array}$$

The following is the simplified expression for $\mathbf{Y}_B$.

$$\mathbf{Y}_B = 0.2s + \frac{3.4s}{5s^2 + 8}$$

Therefore,

$$C_2 = 0.2 \text{ F}$$

$$\mathbf{Y}_C = \frac{3.4s}{5s^2 + 8}$$

The following is the impedance corresponds to $\mathbf{Y}_C$.

$$\begin{aligned} \mathbf{Z}_C &= \frac{1}{\mathbf{Y}_C} \\ &= \frac{1}{\left(\frac{3.4s}{5s^2 + 8} \right)} \\ &= \frac{5s^2 + 8}{3.4s} \end{aligned}$$

The simplified expression for $\mathbf{Z}_C$ equals $sL_3 + \frac{1}{sC_4}$.

Simplify the fraction $\mathbf{Z}_C$ using long division:

$$\begin{array}{r} 3.4s \big) 5s^2 + 8 \big(1.47s \\ \underline{5s^2} \\ 8 \end{array}$$

The following is the simplified expression for $\mathbf{Z}_C$.

$$\begin{aligned} \mathbf{Z}_C &= 1.47s + \frac{8}{3.4s} \\ &= 1.47s + \frac{1}{\left(\frac{3.4s}{8} \right)} \\ &= 1.47s + \frac{1}{0.425s} \end{aligned}$$

Therefore,

$$L_3 = 1.47 \text{ H}$$

$$C_4 = 0.425 \text{ F}$$

Step 5 of 5

Draw the LC ladder network corresponds to the admittance y_{22} :

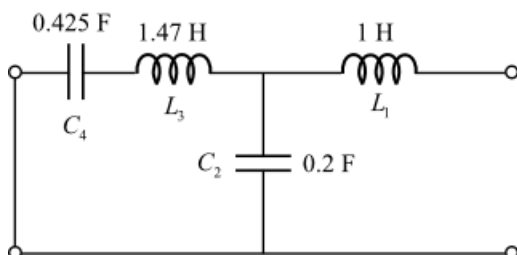


Figure 2